E-commerce User Behaviour Analysis

Prepared by: **Lingzhi Guo**

Executive Summary

In this part of the project I bring together the findings of my behavioural analysis and translate them into concrete operating decisions. My work follows one logic: **portrait of the overall picture, diagnosis of the causes, then targeted action**. Three issues stand out: traffic is not directed at the products that actually sell, the path a user takes before buying strongly affects whether they convert, and the user base is dominated by two segments that need very different treatment.

Key finding 1	Key finding 2	Key finding 3
Traffic and sales are mismatched	Add-to-cart / favourite lifts conversion	91% of users sit in two priority segments

1. Core Conclusions

1.1 Traffic and conversion are misaligned

What I found: the products that attract the most views overlap only weakly with the products that generate the most sales, so overall traffic converts inefficiently.

What it means: the current product display and recommendation logic is not driven by transaction outcomes. A share of exposure is therefore wasted on items that rarely convert.



Figure 1. Schematic: limited overlap between high-traffic and high-sales products.

1.2 The behavioural path shapes conversion

What I found: the journey *browse, then add to cart or favourite, then buy* succeeds far more often than *browse, then buy directly*. Among the intermediate actions, **adding to cart** has the strongest influence on the final purchase.

What it means: cart and favourite actions are clear signals of deeper purchase intent. Encouraging them lowers hesitation and moves users closer to checkout.

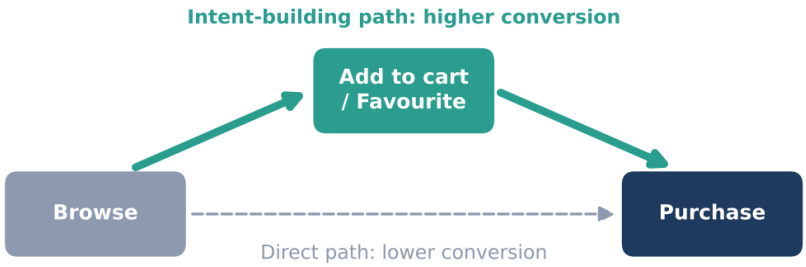
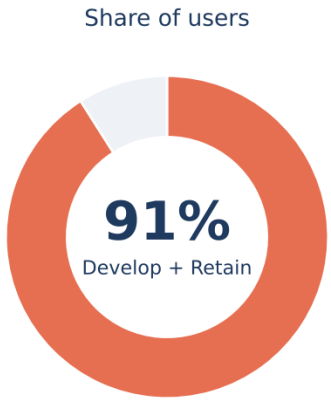


Figure 2. Schematic: the intent-building path outperforms the direct path.

1.3 The user base is polarised, with retention and churn risks side by side

What I found: *Important Develop* and *Important Retain-at-risk (Win-back)* users together make up **91%** of all users and form the core of the base.

What it means: most users are either recently active but buy infrequently, or have become inactive and are drifting away. The platform lacks a stable core of high-value, repeat customers, so differentiated, segment-based operations are needed.



Why this matters

- **Develop users** show interest but lack a reason to purchase again.
- **Win-back users** carry the highest churn risk.
- Small high-value groups cannot carry growth alone, so the two large segments are the main opportunity.

Figure 3. Combined share of the two priority segments (91%).

2. Recommendations and Action Plan

Each recommendation below answers one of the findings above and is assigned to the team best placed to act on it.

2.1 Product recommendation and traffic allocation (Algorithm & Product teams)

- **Best-seller boost (traffic reallocation):** raise the recommendation weight of historically top-selling categories, such as categories 2735466, 1464116 and 4145813, so traffic flows to items that convert.
- **Bundled promotion (higher basket size):** pair high-converting best-sellers with high-traffic but low-converting new or featured products, gaining exposure and conversion together.

2.2 Interaction design and path activation (Product UI/UX & Operations)

- **UI/UX:** add clearer, more prominent add-to-cart and favourite entry points to reduce friction when saving items of interest.

- **Incentives:** reward the key action with light benefits, for example a no-threshold coupon or a limited sample gift for adding to cart, to amplify the high-converting path.

2.3 Prime-time campaign scheduling (Campaign Operations)

User search and interaction activity peaks between **18:00 and 23:00**. I recommend scheduling flash sales, major campaign pushes and version launches inside this window to maximise the traffic they can capture.

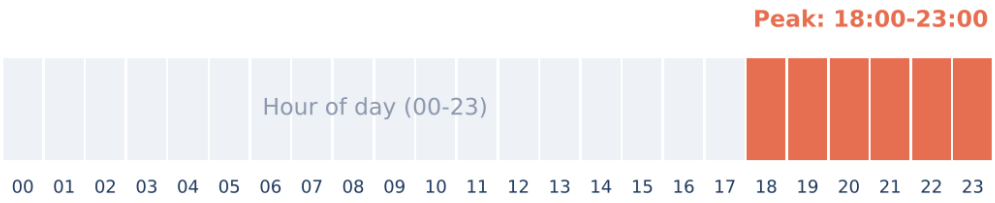


Figure 4. Peak activity window (18:00-23:00) highlighted on a 24-hour timeline.

2.4 RFM-based segmented user operations (User Operations / CRM)



Figure 5. RFM segment map; Develop and Win-back users are the priority groups.

Table 1. Strategy by RFM segment (priority segments shaded).

Segment	Profile	Challenge	Core strategy
Important Value	High frequency, recently active	Costly to maintain; must not be lost	VIP-only benefits, dedicated customer service and privileged experiences to build premium loyalty.
Important Retain	High frequency, recently silent	Loyal but with a short-term gap	Proactive reactivation through app push, email or SMS, plus exclusive repurchase discounts.
Important Develop (priority)	Recently active, low frequency	High interest but no trigger to repurchase	**Raise frequency:** targeted threshold coupons, tiered membership points and precise promotion reminders via CRM to build a buying habit.
Important Win-back (priority)	Recently silent, low frequency	Very high churn risk	**Diagnose then recall:** reward-based surveys to find the cause of churn, combined with a generous comeback offer.

3. Summary

To conclude, I recommend three linked moves: **(1)** align recommendations with what actually sells, **(2)** make add-to-cart and favourite the centre of the conversion path, and **(3)** run differentiated programmes for the two segments that make up 91% of users, concentrating campaigns in the 18:00-23:00 window. Together these target the main sources of lost conversion and lost customers that my analysis identified.